

Week 1 Pre-Class Study Guide: Linear Regression

Time Required: 30 minutes **Complete By:** Before Week 1 live session

Learning Objectives

By completing this pre-class work, you will:

- ☐ Understand what linear regression is and when to use it
 - ☐ Grasp the intuition behind gradient descent
 - ☐ Recognize the sklearn API pattern (fit/predict)
 - ☐ Be familiar with basic terminology (features, target, training, testing)
 - ☐ Come to class ready to code, not learn from scratch
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Required Videos (43 minutes total)

Video 0: StatQuest - Best Fit Line (9 minutes)

Link: <https://www.youtube.com/watch?v=PaFPbb66DxQ>

Video 1: StatQuest - Linear Regression (Stop at 12 minutes 10 secs)

Link: <https://www.youtube.com/watch?v=7ArmBVF2dCs>

What to watch for:

- 0:00-2:00: What is linear regression? (The “line of best fit”)
- 2:00-5:00: Least squares method (minimizing squared errors)
- 5:00-8:00: R-squared interpretation
- 8:00-12:00: Multiple regression (more than one feature)

Key Questions to Answer While Watching:

1. What does the “best fit line” minimize?
2. What does R^2 tell you?
3. Can linear regression handle multiple features (not just one)?

Your Notes:

[Space for your notes]

Video 2.0: Google ML - Gradient Descent (less math) (2 minutes)

Link: <https://www.youtube.com/watch?v=QoK1nNAURw4>

Video 2.1: StatQuest - Gradient Descent (more math) (22 minutes)

Link: <https://www.youtube.com/watch?v=sDv4f4s2SB8>

Key Questions to Answer While Watching:

1. What is gradient descent trying to find?
2. What happens if the learning rate is too large?

3. What happens if the learning rate is too small?

Your Notes:

[Space for your notes]

Required Reading (8 minutes)

Reading 1: sklearn Linear Models Documentation

Link: https://scikit-learn.org/stable/modules/linear_model.html

What to read: Sections 1.1.1 through 1.1.3 only

Focus on:

- The basic `LinearRegression` class
- `.fit()` and `.predict()` methods
- What coefficients represent
- Skip: Advanced topics like Ridge, Lasso (we'll cover in Week 4)

Skim vs. Read Carefully:

- **Read carefully:** Section 1.1.1 (Ordinary Least Squares)
- **Skim:** Examples and mathematical formulations
- **Skip:** Everything after section 1.1.3

Key Concepts to Extract:

1. How do you create a `LinearRegression` model?
2. What does `.fit(X, y)` do?
3. What does `.predict(X)` return?

Your Notes:

[Space for your notes]

Pre-Class Self-Check Quiz

Test your understanding before class. **Don't worry if you can't answer everything perfectly** - we'll cover it all in detail during the live session!

Question 1

What is the goal of linear regression?

- ☐ a) To classify data into categories
- ☐ b) To find the line that minimizes the sum of squared errors
- ☐ c) To cluster similar data points
- ☐ d) To reduce the number of features

Click to reveal answer

Answer: b) Linear regression finds the line (or hyperplane) that minimizes the sum of squared errors between predictions and actual values.

Question 2

What does $R^2 = 0.8$ mean?

- ☐ a) 80% of predictions are correct
- ☐ b) The model explains 80% of the variance in the data
- ☐ c) 80% of the data was used for training
- ☐ d) There's an 80% error rate

Click to reveal answer

Answer: b) R^2 (coefficient of determination) represents the proportion of variance in the target variable explained by the model. $R^2 = 0.8$ means 80% of variance is explained.

Question 3

In sklearn, what does `model.fit(X_train, y_train)` do?

- ☐ a) Makes predictions
- ☐ b) Trains the model by learning from the training data
- ☐ c) Splits the data into train and test sets
- ☐ d) Evaluates the model's performance

Click to reveal answer

Answer: b) `.fit()` trains the model by learning the best parameters (coefficients) from the training data.

Question 4

What happens if the gradient descent learning rate is too large?

- ☐ a) Training is very slow
- ☐ b) The model might overshoot the minimum and diverge
- ☐ c) The model will always find the global minimum
- ☐ d) Nothing - learning rate doesn't matter

Click to reveal answer

Answer: b) If the learning rate is too large, gradient descent takes steps that are too big, potentially overshooting the minimum and causing the loss to increase or oscillate wildly.

Question 5

Why do we square the errors in “sum of squared errors”?

- ☐ a) To make the math harder
- ☐ b) So positive and negative errors don't cancel out
- ☐ c) Because sklearn requires it
- ☐ d) To increase model accuracy

Click to reveal answer

Answer: b) Squaring errors ensures they're all positive, preventing positive errors from canceling out negative ones. It also penalizes large errors more than small ones.

Score Your Pre-Class Quiz

How many did you get right?

- **5/5:** Excellent! You're well-prepared.
- **3-4/5:** Good foundation. Pay extra attention during class to concepts you missed.
- **1-2/5:** That's okay! The videos and reading are dense. The live session will clarify everything.
- **0/5:** No problem! Come to class ready to learn. The instructor will explain from scratch.

Remember: This quiz is for YOU to gauge your preparation, not for a grade!

What to Expect in the Live Session

Segment 1 (0:00-0:20): Welcome & Course Overview

- You'll learn where Week 1 fits in the 8-week journey
- The "big picture" of machine learning

Segment 2 (0:20-0:50): The ML Pipeline

- Step-by-step workflow used in ALL ML projects
- Critical concept: Why we split data before preprocessing

Segment 3 (0:50-1:20): Linear Regression Deep Dive

- Building on what you watched in StatQuest videos
- Loss functions, gradient descent, intuitive explanations (no heavy math!)

BREAK (1:20-1:35)

Segment 4 (1:35-2:15): Live Coding Demo

- **You'll follow along and RUN code** (not type from scratch!)
- California Housing dataset
- Complete pipeline: load → split → train → evaluate → visualize

Segment 5 (2:15-2:45): Pair Programming

- Your turn! Apply what you learned to a new dataset
- Work with a partner

Segment 6 (2:45-3:00): Wrap-up & Homework Preview

What to Bring to Class

Required:

- ☐ Laptop with Python and Jupyter installed (or Google Colab access)
- ☐ This study guide (your notes)
- ☐ Curiosity and questions!

Optional but helpful:

- ☐ Notebook/paper for additional notes
 - ☐ Second screen (if virtual) to have docs open alongside code
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Tips for Success

Before the Live Session

1. Watch videos in order (Linear Regression, then Gradient Descent)
2. Take notes while watching - pause and rewind as needed
3. Skim the sklearn docs (don't need to memorize!)
4. Complete the self-check quiz
5. Prepare your environment (test that Jupyter/Colab works)

During the Live Session

1. Focus on understanding concepts during demos (you won't type during live coding)
2. Ask questions when confused - if you're confused, others probably are too!
3. Take notes on patterns you see
4. Engage actively during pair programming

After the Live Session

1. Complete the post-class homework (30 min)
 2. Review concepts that were unclear
 3. Compare your solution to the provided solution notebook
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Additional Resources (Optional)

If you want to dive deeper BEFORE class (completely optional):

Interactive Tutorials:

- Google ML Crash Course - Linear Regression: <https://developers.google.com/machine-learning/crash-course/descending-into-ml/video-lecture>
- Kaggle Learn - Intro to ML: <https://www.kaggle.com/learn/intro-to-machine-learning>

Visual Explanations:

- 3Blue1Brown - Gradient Descent: <https://www.youtube.com/watch?v=IHZwWFHWa-w>
- Seeing Theory - Regression: <https://seeing-theory.brown.edu/regression-analysis/index.html>

Reading:

- Appendix A: ML Pipeline Template - Preview the code structure you'll use
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Pre-Class Completion Checklist

Before the live session, make sure you:

- ☐ Watched StatQuest Best Fit Line (9 min)
- ☐ Watched StatQuest Linear Regression (12 min)
- ☐ Watched StatQuest Gradient Descent (22 min)
- ☐ Watched Google ML Gradient Descent (2 min)
- ☐ Skimmed sklearn Linear Models docs (sections 1.1.1-1.1.3)

- ☐ Completed the 5-question self-check quiz
- ☐ Tested your Python/Jupyter environment
- ☐ Reviewed learning objectives

Estimated total time: 30 minutes

Questions Before Class?

If anything is unclear, don't stress! The live session will bring it all together.

See you in class!